

# OCR Model Comparison for Egyptian ID Data Extraction

This report evaluates 10 state-of-the-art vision-language models for the task of Optical Character Recognition (OCR) on Egyptian National ID cards, particularly focusing on Arabic text extraction and numeric accuracy (14-digit national ID).

## Best Resolution for Pictures

The optimal image resolution for ID card OCR is 1024x768 pixels (or max dimension of 1024 pixels maintaining aspect ratio). This resolution guarantees that small Arabic diacritics and the 14-digit national ID are sharp enough for the model to read without exceeding the API's maximum image token limit or inflating costs.

## 10 Models Comparison & Pricing (Estimated for 1000 images)

| Model                   | Est. Cost/1000 | Accuracy (Arabic OCR) | Pros                              |
|-------------------------|----------------|-----------------------|-----------------------------------|
| 1. GPT-4o (OpenAI)      | \$13.50        | Excellent             | Highest accuracy, great Arabic    |
| 2. GPT-4o-mini (OpenAI) | \$0.28         | Very Good             | Best cost-to-performance          |
| 3. Llama-3.2-90B-Vision | \$2.50         | Great                 | Best on Groq, extremely fast      |
| 4. Llama-3.2-11B-Vision | \$0.40         | Good                  | Very fast, cost-effective on Groq |
| 5. Claude 3.5 Sonnet    | \$10.00        | Excellent             | Top-tier accuracy and formatting  |
| 6. Claude 3 Haiku       | \$1.00         | Good                  | Fast Anthropic model              |
| 7. Gemini 1.5 Pro       | \$7.00         | Excellent             | Great context handling            |
| 8. Gemini 1.5 Flash     | \$0.15         | Very Good             | Cheapest Tier 1 model             |
| 9. Qwen-VL-Max          | \$3.00         | Very Good             | Strong multi-lingual support      |
| 10. Pixtral 12B         | \$0.50         | Good                  | Good open-weight alternative      |

## Best Model Recommendation

Overall Best Accuracy: GPT-4o (OpenAI) offers the highest precision for complex Arabic OCR tasks.

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Best Cost-to-Performance: Gemini 1.5 Flash or GPT-4o-mini are incredibly cheap while maintaining high accuracy.

Best for Current App (Groq Stack): Llama-3.2-90B-Vision-Preview is the best model currently available on the Groq API. It delivers extremely fast inference speeds, which is ideal for real-time applications, and has capable vision parameters for OCR.

Action Taken: The application has been updated to use 'llama-3.2-90b-vision-preview' to provide the best possible performance while seamlessly integrating with the existing Groq infrastructure.